

DIFFERENCES BETWEEN THE SENSES OF MOVEMENT AND POSITION SHOWN BY THE EFFECTS OF LOADING AND VIBRATION OF MUSCLES IN MAN

D. I. McCLOSKEY

School of Physiology and Pharmacology, University of New South Wales, Kensington, Sydney, N.S.W. 2033 (Australia)

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SUMMARY

Recent experiments have shown that if a muscle is vibrated at 100 Hz with a physiotherapy vibrator, then the joint about which that muscle operates is perceived to be moving as if the vibrated muscle were lengthening. Also, the position of the joint at any moment is perceived to be as if the vibrated muscle were stretched. Here, these distortions of sensation were investigated further at the elbow joint. The velocity of the illusory movement induced by vibration is slowed in proportion to the load borne by tensing the vibrated muscle, and at any load the velocity is slower if the muscle is fatigued. The error of position is increased by loading the vibrated muscle. Vibration of lower frequency but greater amplitude can induce errors of position without inducing illusions of movement. It is argued that separate lines of information can arise in muscle to signal positions and movements.

INTRODUCTION

Recent experiments have shown that signals from receptors in muscle can contribute to position sense or kinaesthesia. These experiments were of two types. First, it was shown that appreciable kinaesthesia persists in joints of the fingers and thumb after the joints themselves are anaesthetized, but the afferents from the remote flexor and extensor muscles which move those joints are preserved^{6,9}. Second, kinaesthetic illusions arising from the stimulation of intramuscular receptors by vibration were described⁷⁻⁹.

Which intramuscular receptors serve position sense is not known. From their experiments using muscle vibration Goodwin *et al.*⁷⁻⁹ argued that the muscle spindle primary afferents are involved. They used a small physiotherapy vibrator oscillating at

100 Hz, applied to the skin overlying muscle tendons, and produced in their subjects proprioceptive illusions in the joints at which the vibrated muscles operate. These were principally illusions of *movement*, the illusory movement being in a direction as if the vibrated muscle were being stretched. Because similar vibration is known from animal experiments to be a powerful stimulus to the primary endings of the muscle spindles², and because muscle vibration in man causes an involuntary tonic reflex contraction generally attributed to excitation of the spindle primaries^{4,10}, Goodwin *et al.* suggested that the vibration-induced illusions of movement they described were due to excitation of these endings. However, primary afferents discharge in the course of normal voluntary muscular activity¹⁵ and no proprioceptive illusions result. Thus, Goodwin *et al.* were led to postulate further that only those discharges from the spindle primaries that are inappropriate for the intended muscular contraction are perceived. In support of this view they reported that the illusion of extension of the elbow joint which usually occurred when the biceps brachii was vibrated could be abolished if the vibration was applied while the biceps was contracting isometrically to exert a large tension. The discharge rate of the muscle spindle primaries is increased when a muscle exerts an increased tension¹⁵, and this increased discharge is appropriate for the intended contraction. Vibration presumably will have little effect on the frequency of firing if this is already high¹³. Thus, vibration of a muscle which is sustaining an increased tension will not cause as much inappropriate discharge of the spindle primary endings as will vibration of an unloaded muscle.

While the predominant effect of vibration is to produce an illusion of *movement* such as would occur if the vibrated muscle were lengthening continuously, there is also an illusion of false *position* associated with vibration. A joint is perceived to be in a position it would take up if the vibrated muscle were stretched^{8,9}. It is possible that illusions of false positions arise from the mathematical integration of signals of velocity. Alternatively, separate lines of information on movement and position may originate in muscle. Goodwin *et al.* recognised these two possibilities, but could not distinguish between them.

The present study extends the approach of Goodwin *et al.*⁶⁻⁹. The distortions of position sense produced in normal human subjects by loading and by vibration are more completely described. Vibration of muscles with a physiotherapy massager at 100 Hz, as described by Goodwin *et al.*, was used here. In some experiments a grosser form of vibration with greater amplitude and lower frequency was also used in an attempt to activate a greater variety of intramuscular receptor types. The results indicate that separate lines of information for movement and for position exist.

METHODS

Experiments were done on normal male and female subjects. None was aware of the various hypotheses being tested.

Two separate types of experiment were done: one was concerned principally with sensations of movement, the other with the sense of position.

Movement

During vibration of the biceps brachii there occurs an illusion that the elbow joint is moving into extension⁷⁻⁹. In the present experiments subjects were asked to give an objective measure of the angular velocity of this illusory movement in the following way. They were blindfolded, and sat with their elbows resting on a support before them and their forearms free to move in a vertical plane. The tendon of the biceps of one arm, designated the experimental arm, was vibrated, and that arm was gently restrained by the experimenter from flexing under the involuntary tonic vibration reflex. The subject was instructed to attend to the velocity of the illusory movement, and then to duplicate this velocity with a voluntary movement of the opposite arm, designated the indicator arm. Subjects were given a short period of training (about 5-10 min) to accustom themselves to the sensations produced by vibration, and to the paradoxical situation in which illusions of movement were experienced of a joint known to be still.

The subjects were required to indicate the angular velocity of the illusory movement during progressive loading of the vibrated biceps. For this, the elbows of both arms were flexed to about 40° from full extension, and loads were added to a bucket suspended over a pulley and supported from a band around the wrist of the experimental arm. Loads were added in increments of 3 lb. The subjects were required to support the loads for no longer than 30 sec at a time, and recovery periods were allowed between observations so as to prevent fatigue in the tested muscle. At the conclusion of these observations the effects of fatiguing the tested muscle were observed. For this, the subjects were required to support a heavy weight (say, 18 lb) in the load bucket for some minutes until they judged that the biceps was considerably fatigued. Then, a series of observations on the velocity of the vibration-induced illusory movement was made while the muscle was unloaded in steps of 3 lb, but without recovery periods in between, so that a state of fatigue was maintained in the test muscle.

For all the above experiments a string was attached to the wrist of the indicator arm and passed to a light lever which turned a potentiometer. This enabled recording of movement at the elbow joint. Vibration was applied with a 'Vibratory Massager' (Pifco Ltd.), oscillating at 100 Hz, with an amplitude of about 1 mm.

Position

Position sense was tested at the elbow joint. Subjects again were blindfolded and sat with their elbows supported before them on a table, and with their forearms free to move in a vertical plane. A perspex screen, marked out in degrees at the elbow, was placed between the arms. One arm, designated the experimental arm, was moved by the experimenter so that the tip of its index finger came to a chosen position on the screen. Ten seconds later the subject was instructed to touch his fingertips together, whereupon he would move his indicator arm by flexing its elbow, and attempt to touch the tips of his index fingers together. Subjects were allowed to take as long over this alignment as they pleased, and there was no question of their having to move into the final position in a single action. The elbow angle chosen by the subject for the

indicator arm was noted. Both fingertips were marked with a fine lengthwise line to allow accurate positioning by the experimenter and noting of the chosen angle against the screen. Usually both arms lay horizontally on the table between trials. Subjects were told that a range of positions for testing would be used between about 20° and 60° into flexion from this horizontal rest position: in fact, only one central position, usually 40°–50° from the rest position, was tested. This central position was chosen to avoid certain accessory clues available nearer full flexion or extension (*e.g.* the weight of the arm, or the distances moved to return both arms to the rest position). Moreover, it was not possible to apply the vibrator to the biceps tendon of an excessively flexed arm. Subjects were not disturbed if successive positions tested were apparently different, because they expected a range of positions to be used.

Each experiment involved a variety of interventions. These were: (1) loading of the experimental arm with weights suspended over a pulley, as described for the movement tests; (2) vibration of the tendon of the biceps of the experimental arm at 100 Hz with the physiotherapy vibrator, also described above; (3) vibration of the tendon of biceps on the experimental side with a padded plunger on a jig-saw attachment of an ordinary workshop electric drill; and (4) combinations of loading and vibration. After a control trial, each intervention under test was used in turn, and then the whole series was repeated until 10 trials of each type were completed.

The vibration with the jig-saw was of a greater amplitude and a lower frequency than that with the physiotherapy massager. The amplitude of the jig-saw's vibration was about 1.5 cm, although when held by the experimenter so as to vibrate the muscle only part of this excursion (perhaps 0.5 cm) was made in contact with the muscle. Thus it delivered more a series of small blows to the muscle tendon than a pure sinusoidal deformation. The frequency of the jig-saw could be varied from 48 Hz to about 2–3 Hz by adjusting the power supplied to it through a variable voltage power supply.

When loads were used the subject lifted the load (either 12 or 15 lb) from the rest position to the point to which the experimenter guided his hand, and held it there until the trial was over: on return to the rest position the arm was unloaded. Vibration was applied to the tendon of the biceps just before the experimental arm was guided to the screen, and remained throughout the trial. At the conclusion of all trials the experimenter guided the experimental arm back to the rest position and the subject was instructed to flex the indicator arm slightly before resting it: this procedure helped obscure any differences in alignment that may have become apparent to a subject detecting differences in the distances travelled by the two arms in returning to the rest position.

RESULTS

Movement

(a) *Loading.* Fourteen subjects were tested in experiments in which they were required to match the angular velocity of vibration-induced illusory movements. All were able to perceive the illusions, but 5 of them were unable to do this reliably on

every application of vibration. The results reported, therefore, are those obtained on the 9 subjects who had no difficulty in perceiving the illusions in the paradoxical situation of the experiment in which the joint was known to be still.

In all 9 subjects, loading the experimental arm caused the illusory movement to be slowed (see Fig. 1). For all 9 subjects the correlation coefficients between velocity and load showed highly significant negative correlations ($P < 0.001$ by Student's t -test). There was, however, a wide variation between subjects, in the velocities perceived and in the slopes of the velocity-load relationships. Straight lines were fitted for the data from each subject by the least squares method. Perceived angular velocities for unloaded muscle ranged from 7 to 30°/sec, the slopes of the velocity-load lines ranged from -0.45 to $-3.1^\circ/\text{sec}/\text{lb}$.

(b) *Fatigue.* The method used to produce muscular fatigue was simple but depended upon the subjects' own assessments of the extent of the fatigue. Consequently, some subjects were tested in a greatly fatigued state, and some were considerably less fatigued. Moreover, observations were made while the fatiguing process continued. Thus, there was considerable scatter in the observations.

In 6 of the 9 subjects, however, fatigue caused a significant slowing of the

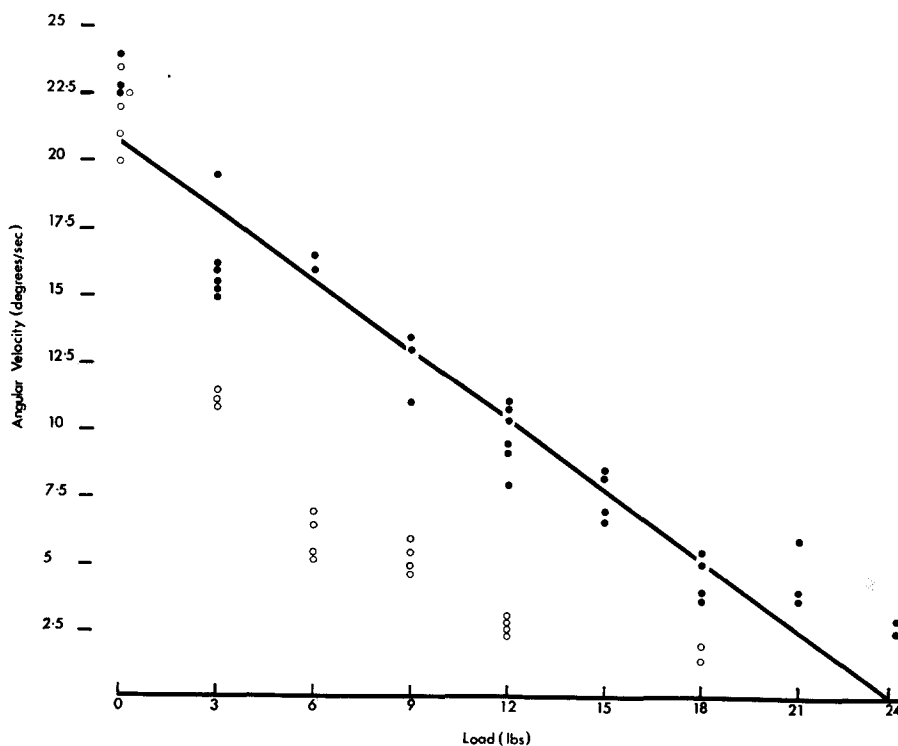


Fig. 1. The angular velocity of the illusory movement of the elbow joint into extension, induced by vibration of biceps brachii at 100 Hz, is plotted against the load at the wrist borne by tensing biceps. The closed circles show the control points, and the line drawn is the line of best fit drawn by the least squares method. The open circles are points obtained when biceps was fatigued through a prolonged period of weight-bearing.

angular velocity of the illusory movements at all added loads. The test of significance was simply that all the points collected during loading in fatigue fell below a line displaced two standard errors of the estimate below the line of best fit drawn through the previously collected velocity-load points (95% of points in the unfatigued velocity-load category would be expected to fall above this line). Fig. 1 shows the results obtained in one subject. Although it appears from Fig. 1 that during fatigue the velocity-load points themselves fall upon a straight line of negative slope, this was not so in all subjects, possibly because of the changing state of fatigue throughout the experiment.

Position

Twenty subjects were used in these experiments. All were normal young adults. Not all the manoeuvres were tried on all subjects. There was considerable variation in the accuracy of these subjects in judging the elbow angle required to bring their fingertips together in the tests described. The positions chosen by different subjects when no intervention was used ranged from 7.5° too flexed, to 7° too extended (means of 10 trials). Consequently, the effects of loading and vibration are described with reference to the control position in each subject individually.

(a) *Loading*. Loading a muscle causes a subject to believe that the joint is in the position it would take up if the loaded muscle were extended. The results of loading the biceps in 15 subjects are shown in Table I. Similar effects were produced in 5 subjects upon loading triceps, except that this caused subjects to believe the arm to be too flexed rather than too extended as with biceps loading.

There was no apparent difference in the effects produced by loading when the subject was required to support a load throughout a series of observations so as to cause progressive fatigue in the loaded muscle, and when he supported the load only during a trial, resting between trials. This is shown for one subject in Fig. 2. Five other subjects gave similar results.

(b) *Vibration at 100 Hz*. Vibration with a physiotherapy massaging device at 100 Hz caused subjects to believe that a vibrated muscle was more extended than it was. The results for subjects in whom biceps was vibrated are shown in Table I. Similarly significant errors in the opposite direction were found in 4 subjects in whom the triceps was vibrated.

In further trials on 10 subjects different durations of vibration were used. These subjects were required to attempt to approximate their fingertips in separate trials at 5 or 10 sec after the experimental arm had been positioned and the vibration begun. In all 10 subjects the vibration produced significant errors of the kind described above. In none of the subjects was there a significant difference between the error made after 5 sec vibration and that made after 10 sec vibration.

(c) *Loading and vibration (100 Hz)*. When the biceps was both loaded and vibrated a subject believed that the arm was more extended than it was. Moreover, the errors induced by loading and vibrating together were greater than those induced by vibration alone (see Table I).

There was one notable exception to this. One subject had a particularly strong

TABLE I

POSITION SENSE AT THE RIGHT ELBOW JOINT, TESTED AT 50° FROM FULL EXTENSION

Positions are shown for the elbow angles chosen for the indicator arm in attempted matches. Angles are shown in degrees ($\pm$ standard error of mean for 10 trials) of error from a true match. Errors in extension are shown as positive, errors in flexion are negative. Angular velocity (mean of 5 trials) of illusion of movement induced by vibration is shown for some subjects. Significance levels come from *t*-tests. n.s. = not significant. Note that one subject, marked with an asterisk, gave results different from the others (see text).

Subject	Control (°)	Load biceps (°)	Load vs. control (<i>P</i> <)	Vibrate biceps at 100 Hz (°)	Vibration vs. control (<i>P</i> <)	Angular velocity (°/sec)	Vibration + load (°)	Vibration + load vs. vibration (<i>P</i> <)
C.G.	-0.8 $\pm$ 0.30	+1.6 $\pm$ 0.56	0.01	+6.0 $\pm$ 1.02	0.001	2.5	+ 4.9 $\pm$ 1.17	n.s.
P.E.	-4.5 $\pm$ 0.50	+0.9 $\pm$ 0.60	0.001	+0.9 $\pm$ 0.37	0.001	1.5	+ 7.7 $\pm$ 0.87	0.001
I.N.	-0.1 $\pm$ 0.60	+6.6 $\pm$ 1.15	0.001	+8.8 $\pm$ 0.89	0.001	6.5	+12.9 $\pm$ 0.54	0.001
P.C.	-0.4 $\pm$ 0.51	+4.3 $\pm$ 0.14	0.001	+3.1 $\pm$ 1.30	0.01	3.0	+10.9 $\pm$ 0.91	0.001
M.S.	-3.3 $\pm$ 0.44	+1.6 $\pm$ 0.50	0.001	-1.0 $\pm$ 0.46	0.01	8.5	+ 3.6 $\pm$ 0.37	0.001
J.C.	-7.5 $\pm$ 0.67	+0.3 $\pm$ 0.67	0.001	+6.4 $\pm$ 0.68	0.001	12.5	+11.9 $\pm$ 0.34	0.001
E.M.	-3.8 $\pm$ 0.67	+5.0 $\pm$ 1.22	0.001	+5.2 $\pm$ 1.41	0.001	10.0	+ 7.8 $\pm$ 1.26	n.s.
R.M.	+0.7 $\pm$ 0.46	+1.9 $\pm$ 0.72	n.s.	+3.3 $\pm$ 0.90	0.01	4.0	+ 6.6 $\pm$ 0.56	0.01
A.L.	-5.4 $\pm$ 0.63	+0.6 $\pm$ 0.40	0.001	+4.3 $\pm$ 0.51	0.001	22.5	+ 4.7 $\pm$ 1.03	n.s.
J.M.	-2.7 $\pm$ 0.47	+0.5 $\pm$ 0.22	0.001	+4.8 $\pm$ 0.78	0.001	35.0	+ 3.7 $\pm$ 0.65	n.s.
I.C.	-2.7 $\pm$ 0.64	+0.7 $\pm$ 0.52	n.s.	+0.2 $\pm$ 0.55	n.s.	1.0	+ 7.0 $\pm$ 0.48	0.001
R.H.	+2.3 $\pm$ 0.72	+5.3 $\pm$ 0.32	0.001	+2.1 $\pm$ 0.45	n.s.	—	+ 5.8 $\pm$ 0.37	0.001
M.P.	+1.9 $\pm$ 0.72	+3.4 $\pm$ 0.89	n.s.	+4.0 $\pm$ 0.78	0.05	—	+ 7.9 $\pm$ 0.41	0.001
R.B.	0.0 $\pm$ 0.64	+1.4 $\pm$ 0.92	n.s.	+3.2 $\pm$ 0.32	0.001	—	+ 5.8 $\pm$ 0.24	0.001
M.J.*	+1.2 $\pm$ 0.64	+3.4 $\pm$ 0.98	n.s.	+23.6 $\pm$ 1.54	0.001	18.5	+ 6.1 $\pm$ 0.98	0.001

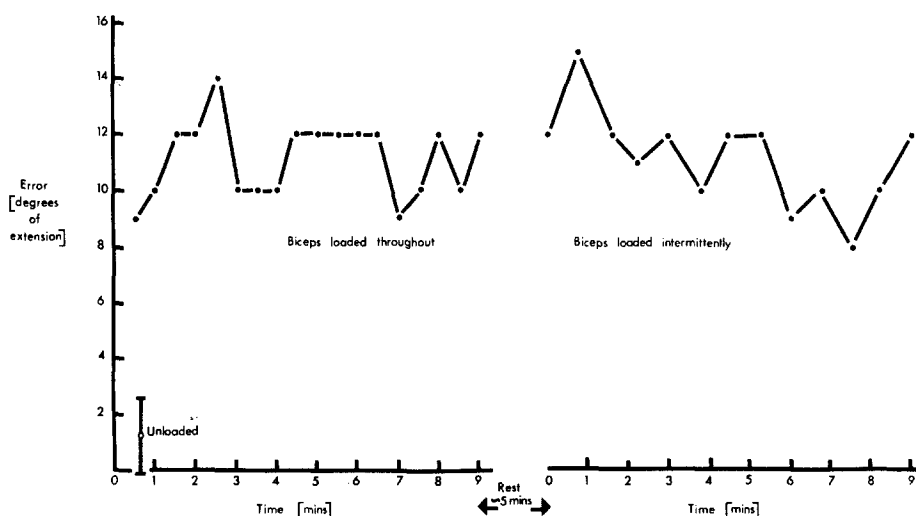


Fig. 2. The error of the estimate of position of the partly flexed elbow joint, in degrees of extension, is plotted against time. On the left of the figure the estimates were made while the biceps of the experimental arm was tensed throughout to support a load of 12 lb applied at the wrist: this caused subjective fatigue, and the subject could not support the load beyond the period shown. On the right of the figure the estimates were made, after an intervening rest period, over a period similar to that used in the first series, but in this series the subject supported the load only intermittently and rested between estimates. At bottom left is shown the subject's control estimate of the position of the unloaded elbow joint (mean of 10 trials $\pm$ 2 S.E.M.).

illusion of movement in response to vibration at 100 Hz. This subject continued to experience this illusion undiminished when the finger of the experimental arm was placed against the perspex screen, although other subjects found the movement illusion much less intrusive once the experimental arm was so evidently anchored. This single subject, marked with an asterisk in Table I, made significantly smaller errors of position when the vibrated arm was both loaded and vibrated, than when it was simply vibrated.

(d) *Lower frequency vibration.* Vibration with the jig-saw, like vibration with the physiotherapy massager, caused subjects to believe that their elbows were positioned such that the vibrated muscle, whether biceps or triceps, was stretched.

Usually, the jig-saw vibrator, with a frequency of oscillation only half that of the physiotherapy vibrator, caused illusions of movement. These were detected by a procedure similar to the movement tests described above. A subject was asked to keep forearms aligned and then the biceps of the experimental arm was vibrated but restrained from moving. An illusion was shown to be present when the subject moved the indicator arm into extension. By varying the voltage supply to the jig-saw it was possible to slow the frequency of its oscillation. This was done in a series of steps, until application of the slowed vibration no longer caused an illusion of movement. The criterion for this was that the subject did not move the indicator arm into extension on 10 successive trials of vibration.

When a muscle was vibrated at a frequency at which no illusion of movement

TABLE II

POSITION SENSE AT THE RIGHT ELBOW JOINT, TESTED AT 50° FROM FULL EXTENSION

Positions are shown for the elbow angles chosen for the indicator arm in attempted matches. Angles are shown in degrees ($\pm$ standard error of mean for 10 trials) of error from true match. Errors in extension are shown as positive, errors in flexion are negative. The frequency of jig-saw vibration just low enough to cause no perceived illusion of movement is shown. Significance levels were obtained by *t*-tests. n.s. = not significant.

Subject	Control (°)	Biceps vibration at 100 Hz (°)	Biceps vibration with jig-saw (°)	Frequency of jig-saw (Hz)	Jig-saw vs. control (<i>P</i> <)	Jig-saw vs. vibration at 100 Hz (<i>P</i> <)	Jig-saw vibration + load (°)	(Jig-saw + load) vs. jig-saw (<i>P</i> <)
P.E.	+1.1 $\pm$ 0.62	+ 4.1 $\pm$ 0.64	+ 6.9 $\pm$ 0.80	35	0.001	0.01	+ 4.2 $\pm$ 0.81	0.05
I.C.	-2.7 $\pm$ 0.64	+ 0.2 $\pm$ 0.55	+ 6.7 $\pm$ 0.83	35	0.001	0.001	+ 7.0 $\pm$ 0.48	n.s.
M.P.	+0.9 $\pm$ 0.68	+ 3.8 $\pm$ 0.62	+11.7 $\pm$ 0.64	30	0.001	0.001	+ 5.4 $\pm$ 0.51	0.001
R.B.	0.0 $\pm$ 0.64	+ 3.2 $\pm$ 0.32	+11.4 $\pm$ 1.20	30	0.001	0.001	+ 5.8 $\pm$ 0.69	0.001
R.H.	+2.3 $\pm$ 0.72	+ 2.1 $\pm$ 0.45	+10.1 $\pm$ 1.17	35	0.001	0.001	+ 6.2 $\pm$ 0.72	0.01
S.G.	+7.0 $\pm$ 0.90	+ 8.9 $\pm$ 0.74	+12.0 $\pm$ 0.84	25	0.001	0.01	+13.2 $\pm$ 0.84	n.s.
M.S.	-4.0 $\pm$ 0.76	- 0.7 $\pm$ 0.60	+ 2.2 $\pm$ 0.56	30	0.001	0.001	+ 1.8 $\pm$ 0.64	n.s.
J.R.	+2.2 $\pm$ 0.67	+ 7.0 $\pm$ 0.98	+ 5.4 $\pm$ 1.24	20	n.s.	n.s.	—	—
J.M.	-3.2 $\pm$ 0.52	+ 1.6 $\pm$ 0.68	+ 9.7 $\pm$ 0.83	30	0.001	0.001	+ 4.6 $\pm$ 0.72	0.001
I.N.	-1.2 $\pm$ 0.42	+ 5.6 $\pm$ 0.76	+16.1 $\pm$ 0.83	35	0.001	0.001	+11.0 $\pm$ 0.88	0.001
T.H.	+3.4 $\pm$ 1.04	+14.8 $\pm$ 0.75	+12.7 $\pm$ 0.97	30	0.001	n.s.	+11.2 $\pm$ 0.76	n.s.

was perceived, subjects believed that the joint was in a position such as it would take up if the vibrated muscle were stretched. This was true for vibration of biceps or triceps. Moreover, the error of position made during this low-frequency vibration was usually greater than the error made by the same subject when 100 Hz vibration was used (see Table II). Alternatively, the lower frequency jig-saw vibration could be chosen to give a movement illusion of a velocity comparable to that induced by the 100 Hz physiotherapy vibrator: when this was done the error of position made greatly exceeded the error induced by the 100 Hz vibration. This alternative method was sometimes difficult to use because the movement illusions induced by the jig-saw vibration, even at the maximum frequency attainable with it (48 Hz), were slower than those induced by the 100 Hz physiotherapy vibrator.

(e) *Lower frequency vibration plus loading.* It has been described above that vibration at 100 Hz causes subjects to make greater errors of position when it is applied to a loaded muscle than when it is applied to an unloaded muscle. A similar effect was not usually found when lower frequency vibration, which caused no movement illusions, was used. Loading either reduced the errors of position, or had no significant effect upon them (see Table II).

DISCUSSION

The present experiments support and extend the work of Goodwin *et al.*⁶⁻⁹ on the contribution of intramuscular receptors to position sense, or kinaesthesia.

Goodwin *et al.* described illusions of movement induced by vibration of muscles at 100 Hz. The experiments described here provide evidence that the illusions of position, also described by Goodwin *et al.*, do not simply result from mathematical integration by the central nervous system of discharges signalling movement. First, illusory movements induced by vibration are usually too fast for their simple integration to result in position errors of the size of those found here. Second, there is no demonstrable difference in the errors of position which result from applying vibration to a muscle for 5 and 10 sec. Third, loading a muscle slows the illusory movement but increases the vibration-induced error of position. And, fourth, a lower frequency of vibration with increased amplitude can be chosen which will give no perceived illusions of movement, although it will still cause errors of position.

It follows from the considerations above that signals of position can arise which exist apart from any which arise from integration of velocity signals — that is, that separate lines of information are available for signalling position and movement. It remains possible that some integration of movement signals is also sometimes used in the final computation of position, although its influence was over-riden here. This is, for example, suggested by the exceptional result found in the one subject in whom loading a muscle reduced the position error induced by vibration. Apart from this, in the conditions of the present study, integrations of movement signals were less effective than separate signals more specifically concerning position.

By asking subjects to match the angular velocity of vibration-induced illusory movements, it was possible in this study to avoid a problem experienced by Goodwin

*et al.*⁹. They observed the effects of muscle vibration in subjects given specifically a position matching task. Thus, many of their subjects became confused, and hesitated, when they followed a movement illusion into a position which other signals indicated was false. In contrast, subjects in this study were prepared, and in fact expected, to misalign their arms when matching the illusory movement in one with a real movement of the other. While this made it easier to observe properties of the illusory movements, some subjects had difficulty perceiving the illusions reliably enough to do this form of test. All, however, did perceive illusions in most trials. Such difficulty is not unexpected in subjects meeting stimuli given in an unusual way: for example, many untrained subjects have difficulty in making reliable observations on visual after-images. It is not surprising that a wide range of velocity-load relations were found in the different subjects. The mechanical state of the vibrated muscle, the amplitude of vibration actually transmitted through the muscle, and the state of fusimotor facilitation are all factors likely to vary widely from subject to subject.

Loading a muscle reduced the angular velocity of the illusory movement induced by vibration. This fits well with the observations of Goodwin *et al.*⁹ that in a heavily loaded muscle the illusion may be abolished. Further, as pointed out by Goodwin *et al.*, this provides evidence that intramuscular receptors are responsible for the illusory sensations, since only these should have been so markedly affected by the strength of contraction. It also supports their idea that one perceives only that part of the discharge of the intramuscular receptors which would not usually be present at a given level of contraction. This idea is explained more fully above in the Introduction. They noted, however, that 'similar considerations would apply if the tendon organ were to be responsible for the illusion so this particular observation does not of itself discriminate them and spindles as the causative receptor'. Since there is no reason to expect changes in overall tendon organ activation during the increased efforts required in fatigue, the significant changes in the movement illusion seen in fatigue suggest that tendon organs are not the receptors responsible for the illusion.

Cohen³ showed that muscular loading led to greater than usual errors in reproducing positions of the shoulder joint, but these errors were not of a systematic nature. Here, loading was shown to cause subjects to believe that the joint has moved in the direction of loading. This might be caused by the discharge of stretch or tension receptors within the loaded muscle. However, such discharges, being 'appropriate' for the contractions performed, would not, on the arguments outlined above, be expected to contribute to position sense. In favour of this is the observation that progressive fatigue of a loaded muscle does not systematically alter the estimate of the position of its joint (Fig. 2), although it is likely that the discharge from muscle spindles would increase throughout the fatigue. Other intramuscular stretch or tension receptors may be involved, or it might be that a consistent error is made by the nervous system in computing the appropriateness of spindle discharge at any load. Another possibility is that the errors arise when muscles pull upon joint receptors so as to alter their discharge rates at any given position: it is known from animal experiments that muscular action can produce such effects^{12,14}.

Vibration at 100 Hz with a physiotherapy massager produces position errors,

as described by Goodwin *et al.*^{8,9}. Similar effects have been described by Eklund⁵ at the knee joint. In all but one subject, vibration applied to a loaded muscle caused significantly larger errors of position than it did with an unloaded muscle. It is not clear, however, why the errors of position induced by vibration should be increased by loading. Possibly the vibration is transmitted better mechanically through a contracted muscle; or perhaps it is more effective in this state upon a second type of receptor, not responsible for the illusions of movement.

Lower frequency vibration, although producing no movement illusions, was quite effective in producing errors of position. This is further evidence that position is not indicated by the integration of movement signals. This gross form of interference has had no equivalent in animal studies. The experiments of Bianconi and Van der Meulen¹ showed that transverse vibration of a muscle belly in the cat excited both muscle spindle primary and secondary endings. The jig-saw vibration was used here in an attempt to excite a greater variety of intramuscular receptor types, and it is probable that both muscle spindle primary and secondary endings were excited here, together with tendon organ receptors, and other receptors within the muscle, tendons or joints. Any of these receptors might have been responsible for the position errors which persisted when no movement illusions were perceived. If the important difference in the present experiments was that the vibration at 100 Hz selectively excited muscle spindle primary afferents, whereas the grosser vibration excited both spindle primaries and secondaries, then the following observation may be relevant. Jansen and Matthews¹¹ have pointed out that 'while the primary ending signals both the length and rate of change of length of the muscle, the secondary ending signals mainly length with a sensitivity not very different from that of the primary ending. Thus the difference in the frequency of discharge of the two kinds of ending approximately signals the rate of change of length of the muscle'.

It appears from the present study that subjective judgements of the static positions of joints and judgements of movements of joints can use different lines of information. It is suggested that the term 'position sense' be reserved for the static judgements, and 'kinaesthesia' for the dynamic ones, and that the two terms should not be regarded as synonymous.

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